

22 66 53 - Laboratory Chemical-Waste and Vent Piping

All wet laboratories, all fume hoods with cup sinks, and all other locations using chemicals which could damage standard DWV piping in the event of an accidental spill shall be provided with a lab waste and vent system. The lab waste and vent system shall be made of chemically resistant piping materials. The lab waste and vent system shall merge with the building sewer at a location outside of the building. Prior to combining the two systems, provide a lab waste sampling station (typically a manhole with a sump for sample collection).

Acceptable Piping Materials

Above and below grade Lab Waste and Vent Piping; (Alternate material); Chemical resistant coated cast iron pipe with matching mechanical joint DWV fittings. (Duriron or equal).

Above grade Lab Waste and Vent Piping; (Alternate material); Chemical resistant glass acid piping with glass DWV mechanical joint fittings. (Pyrex , Kymax, or equal).

Above and below grade Lab Waste and Vent Piping

Charlotte Pipe STM F 2618: Standard specification for CPVC Pipe and Fittings for Chemical Waste Drainage. Scope: this specification covers the performance requirements of CPVC pipe, fittings, and solvent cements used in chemical waste drainage systems. F493: Specification for solvent cements for CPVC pipe and fittings. Scope: This specification provides requirements for CPVC solvent cement to be used in joining CPVC pipe and socket fittings.

Lab waste vents shall be located a minimum distance of 50 feet horizontally from HVAC air intakes and occupied roof areas. Additionally, lab waste vents shall be extended with a support system to a level seven feet above the roof when the roof is occupied (such as in the case of roof mounted greenhouses).

Lab waste vents shall not be located in confined roof wells where HVAC air intakes are located.

Provide accessible cleanouts at all levels of the building the same as required by code for the first floor and lower levels.

Floor drains shall not be provided in laboratories using hazardous materials as precaution against such materials being discharged to the sewer system. The floor system should be designed to provide containment of such materials in the event of an accidental spill.